

ClassInsight: An agentic AIEd application for responsive teaching

T a l i a B e r l e r

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PROBLEM STATEMENT

Responsive teaching

- Individualized > General
- Real-time > Delayed
- Humanities

Traditional Approaches

- Questions
- Discussion
- Quiz

SOLUTION

ClassInsight



Inputs

Educator provides:

1. Email List
2. Question
3. Model Response
4. Categories



Outputs

2 Main Products:

1. Quantitative: aggregated, anonymous visuals
2. Qualitative: individual feedback



Start Assignment



Live

Real-Time Class Demo

Begin

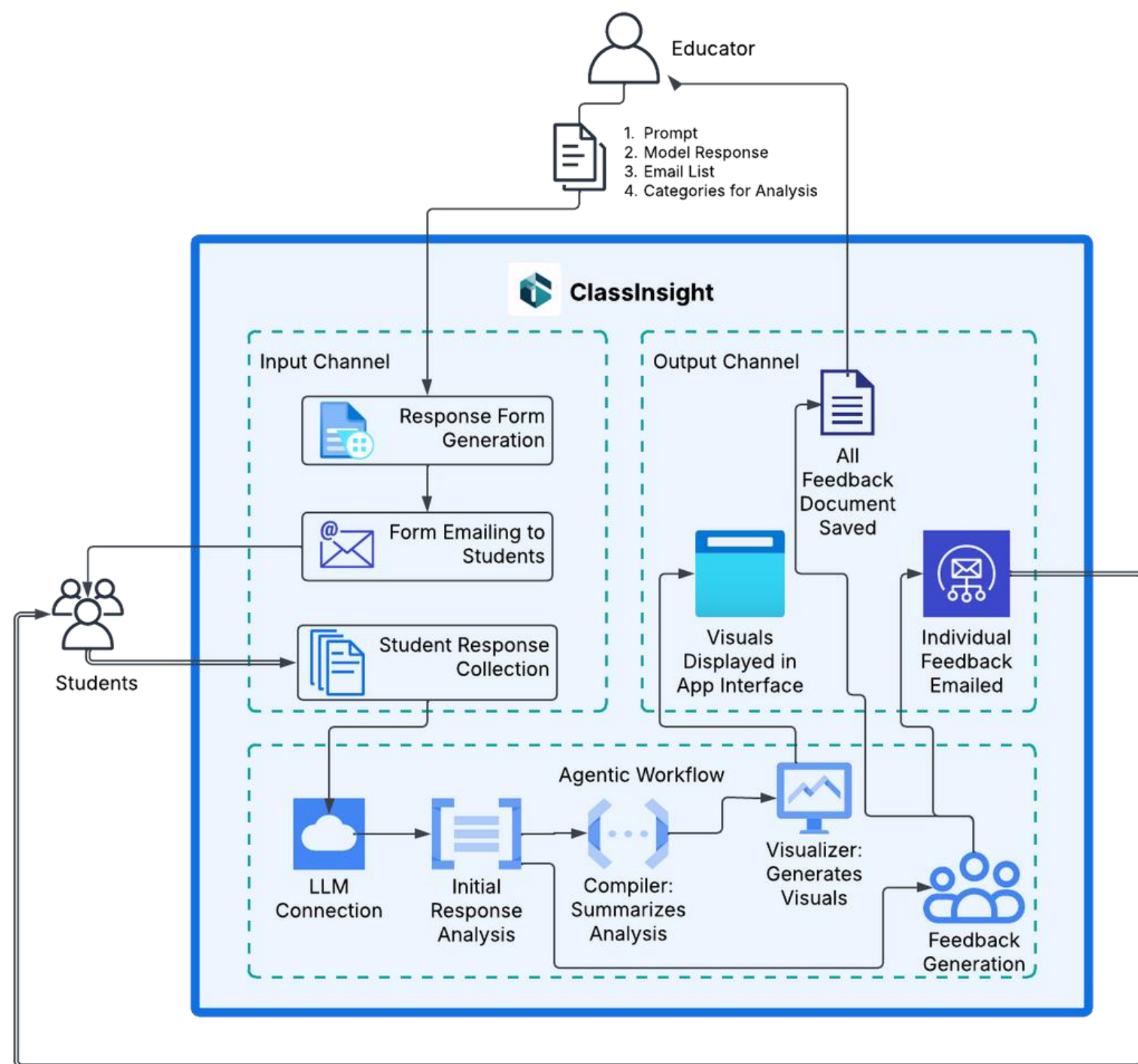


Asynchronous

At-Home Assignments

Begin

METHODOLOGY



```
async def run_analysis(self):
    """Main analysis workflow"""
    try:
        # retrieve/create assistants, save ids
        assistants = self.retrieve_assistant()

        # Initialize assistant IDs
        visualizer_id = None
        compiler_id = None

        # Get IDs by checking assistant names
        for assistant in assistants:
            if assistant.name == "Visualizer":
                visualizer_id = assistant.id
            elif assistant.name == "Compiler":
                compiler_id = assistant.id

        if not visualizer_id or not compiler_id:
            raise Exception("Failed to retrieve both assistant IDs")

        answers = self.RESPONSES

        # Initial analysis
        print("Getting initial analysis batches...")
        insights = await self.process_async_msgs(answers, n=5)
        print(insights)

        # Compile reports
        compiled_reports = await self.compile_insights_helper(insights, final=False, compiler_id=compiler_id)
        if not compiled_reports:
            raise Exception("No valid compilation results received")
        print(compiled_reports)

        # Final aggregation
        done = False
        i = 0
        while len(compiled_reports) > 1 and not done:
            print(f"Combining final compiled reports: batch {i+1}...")
            print(f"Current number of reports to combine: {len(compiled_reports)}")
```



RESULTS

Performance

Accuracy

- Incorrect and partially correct responses: CV $\sim 5\%$
- Correct responses: CV $\sim 50\%$
- Larger batch sizes $\rightarrow$ improved accuracy

Demos

Classrooms and educators

- PETAL conference
 - Batch size: 10 and 25
- UM law classes
 - Batch size: 70 and 80



CONCLUSION & SIGNIFICANCE

- ✿ Real time assessment for responsive teaching
- ✿ Timely feedback improves education
- ✿ ClassInsight empowers educators and students

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Thank you!

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